

The Unmeasured

Six Tellings

A stylized, handwritten-style logo or signature, possibly representing the letters 'GA' or 'SA', rendered in a light gray color.A handwritten signature in a light gray color, appearing to read 'H. W. Anderson'.

I. The Optician's Granddaughter

The woman who conducted the experiment had been raised in her grandfather's shop among trays of unfinished lenses, and she had understood from an early age that light is a thing which can be persuaded. This is why she was unsurprised by the apparatus and profoundly disturbed by its result, which is the opposite of how the other physicists in that department received it.

The arrangement was simple enough to explain to a child: a source, a barrier pierced by two narrow openings, and beyond it a plate to receive whatever arrived. Send light through and the plate recorded not two bright lines, as any reasonable person would predict, but a ladder of bands, bright and dark in alternation, running out to either side and diminishing as it went. This much had been known for a century and had been filed under the behaviour of waves, and nobody had lost any sleep over it.

She altered the experiment until the thing could no longer be dismissed. She reduced the source until it released its light in single units, one at a time, with intervals long enough that no two could possibly have kept company. Each unit arrived at the plate as a single point, a small definite mark in a single definite place, precisely as a particle should. And yet as the marks accumulated across the hours – hundreds, then thousands, arriving alone, arriving separated, arriving with nothing whatever to interfere with – they assembled themselves, patiently and without exception, into the same ladder of bands.

She sat with that plate for a long while. Each mark had been made by one arrival. Each arrival had travelled alone. And the position of each solitary mark had nonetheless been governed by the existence of a second opening that the arrival had not gone through. Cover the second opening and the bands vanish; the marks redistribute themselves into an ordinary blur. The unchosen path had been doing work. It had been contributing to an outcome while having, in the crude sense of the word, never happened at all.

That was the sentence she could never afterwards unlearn, and she was careful to phrase it precisely, because she had watched colleagues ruin themselves by phrasing it loosely. She did not claim the unit had gone through both openings, which is a statement that means nothing. She claimed something narrower and considerably worse: that the alternative which did not occur was not merely absent from the world. It was present in the arithmetic. The paths not taken were participating.

There is a distinction here which she spent the remainder of her career defending against people who found it inconvenient. A coin concealed beneath a cup is already heads or already tails, and the man who does not know which is merely ignorant; his ignorance has no effect whatever upon the coin. What her plate demonstrated was something else entirely, something for which ordinary ignorance is not an explanation. The alternatives were interfering. And that which can interfere is not nothing.

II. What the Looking Costs

The inspector, greyer now and two years from a pension she did not intend to collect, was asked by a younger colleague why she conducted her interviews in a bare room with the recorder placed where the subject could see it. He assumed this was a technique for producing anxiety. She told him it was the opposite; it was an admission. She had come to understand that there is no such thing as observing a man without also changing him, and that she preferred the change to be one she had chosen deliberately rather than one she had introduced by accident and then mistaken for evidence.

She had learned this the way one learns most things worth knowing, which is expensively. Years earlier she had sat across from a boy who arranged his hands too carefully on the table, and she had read that arrangement as guilt when it was in fact terror, and part of what she had failed to account for was that the terror was partly her own manufacture. He had not walked into that room afraid. He had become afraid inside it, of her, of the door, of the recorder, of the particular way an official building smells. She had then read the fear as though it were a property he had carried in with him.

This became, in the last decade of her working life, something close to a doctrine. The act of measurement is not a window. It is a transaction, and the observer pays. A thermometer takes heat from the water it reports on. A man who knows he is being watched is no longer the man who was not being watched, and the difference between the two is not a distortion to be subtracted out; it is a genuine physical event that the observer has caused. There is no vantage from which one looks at a thing without touching it.

Her colleague objected that this made truth impossible. She said it made a certain fantasy of truth impossible — the fantasy of the invisible witness, the perfect glass, the report that costs nothing — and that the disappearance of a fantasy is not the same as the disappearance of knowledge. One simply has to

account for the price. She then went back into the bare room, where the recorder sat in plain view because the young man on the other side of the table deserved to know what he was being changed by.

III. The Surveyor's Stone

The surveyor had spent thirty years fixing points upon the earth, and he understood better than most men how little a point is. He had triangulated a border across four hundred kilometres of scrub, and he knew that every monument he had set into the ground was a lie of a very useful kind: a declaration that a particular place would hold still, made about a crust that does not hold still, on a world that is at every instant somewhere it has never been before.

When his wife died he buried a small box at the base of a marker stone on the ridge above the town, at coordinates he recorded to a precision the task did not require, because precision was the only form of grief available to him. He returned the following year on the same date and found the stone and dug and recovered the box, and he understood, standing there with soil under his nails, that he had not returned to anything.

He worked it out that evening with a pencil, for the pleasure of the arithmetic rather than any need to be convinced. In the year between his two visits the earth had carried that ridge some nine hundred and forty million kilometres around the sun; the sun had carried the whole arrangement several thousand million kilometres further around the wheel of the galaxy; the galaxy itself was falling toward a distant mass at a speed he could estimate but not verify. Nothing in that column of figures had returned to anywhere. He had gone back to a name – the ridge, the stone, the date – and a name is not a place.

The deeper observation came later, and he never put it on paper. It was that the same held for the moment as well as the position. He had imagined the anniversary as a return, and it had not been one. He had come back to a description and found it occupied by an entirely different arrangement of the universe, and the only thing that had persisted between the two visits was a story he was telling himself about continuity – a story that had been useful for standing upright in, and false in every particular.

He continued to set monuments for another decade, and he set them well, and he told his apprentices what he had come to believe: that their trade was not the recording of places but the manufacture of agreements about places, and that a man who forgot the difference would eventually try to go back to something and

be destroyed by discovering that he could not. There is no address one can leave and then re-enter. There is only the description of an address, which anyone may write down, and which the universe is under no obligation to still be standing in.

IV. The Ledger Without a First Entry

The restorer took the commission because the wood interested him and not the mystery, which he considered, at the outset, to be a matter of poor record-keeping. The piece was a writing cabinet from a house in the interior, and inside a false panel there was a manuscript: forty pages, unsigned, on paper he could date within a decade by touch alone, describing in careful detail a method for treating marble against frost that he had learned as a young man from a carver in the hill country.

That was the first difficulty, and it was not yet serious. The carver had been a famous man in a small way, and it was entirely possible that his method had reached this house by some ordinary route. The restorer wrote to the workshop that had inherited the carver's papers and received, in due course, a reply that troubled him considerably. The method appeared in the carver's own notes, undated, in a hand that was not the carver's, on paper of roughly the same decade as the manuscript in the cabinet, with a marginal note indicating that it had been copied from a document received from the house in the interior.

He spent the better part of a year attempting to open the circle. He established that the house had received the document from the workshop; he established that the workshop had received it from the house. Each half of the sequence was documented, credible, and independently confirmed. Together they described a thing which had arrived from itself. Somewhere in the chain there had to be a man who had first thought of the method, and no chain he could construct contained that man. Every route he traced returned him to a copy of a copy, with no original anywhere in it, the whole structure standing upon nothing and standing nonetheless.

He did not find an answer, and he came to suspect that the absence of the answer was not an absence of records but a property of the thing. He set it down finally, in the one letter he wrote on the subject, with the plain formulation that had come to him while planing a surface and thinking of something else: that we assume every real thing must have a first moment, and that this assumption is a habit rather than a law. A closed loop needs no beginning. It

needs only to be consistent with itself at every point along its length, which this one was, ruinously.

What frightened him about it, he wrote, was not the mystery. It was the smoothness. Nothing in the documents looked wrong. Every step was ordinary. Only the whole was impossible, and the whole was not visible from any single step – which meant, he supposed, that a man might spend his life inside an impossibility and never encounter anything but reasonable paperwork.

V. The Lodger

There was a man who took a room above a bakery in the old quarter and who could not be proven wrong about anything. This was his central quality, the thing he was known for in the taverns, and he mistook it for the whole of his intelligence.

He held that the earth might be flat within some higher geometry in which our notion of curvature was an artefact of perspective. He held that water was a variety of stone in a condition we had not learned to recognise, and paper a slow liquid, and that the men who laughed at him were the intellectual descendants of the men who had laughed at the suggestion that the earth was not the centre of anything. Each of his positions was constructed so that no observation could touch it. Each was, in the narrow sense he cared about, unrefuted. He believed this made them strong.

A schoolmaster who drank at the same table tried once, without unkindness, to show him where the error lay. He pointed out that the proposition which cannot be contradicted by any possible observation has purchased that safety at a terrible price: it has given up the ability to tell you anything. A claim earns its standing by forbidding something. The heliocentric model was not a better story than the old one; it was a more dangerous one, because it committed itself in advance to the positions of the planets on nights that had not yet occurred, and could have been executed by a single night's disagreement, and was not. That risk was the whole of its authority.

The lodger heard this as an insult and continued for another twenty years, and his theories multiplied, and not one of them ever came to anything, because none of them could ever be wrong and therefore none of them could ever be found right. He died with a trunk full of notebooks, every page of which remained true in exactly the way that costs nothing.

The schoolmaster, who outlived him, kept one lesson from the acquaintance and repeated it to his students until they were sick of it. That the imagination is enormous, and that therefore what we can imagine is very nearly worthless as a guide to what is so. Anything can be conceived. That is precisely why conceivability was never going to be the filter. The filter is the willingness to say, in advance and in public, which observations would end the matter – and then to go and look.

VI. Ash

In his sixties the restorer burned his father's papers, in the yard, in a brazier meant for cooking, over the course of a long afternoon. There was no drama in the decision. Nobody living could read the tuner's shorthand, the sheets were foxed past use, and he had been carrying the box from house to house for thirty years out of a reluctance he had never examined.

He watched the first sheets go and found himself thinking, with the pedantry of a man who works with materials, about what he was actually doing. He was not destroying anything. Not one atom in that brazier was being removed from the world; the whole of its mass was going into the air as gas and into the tray as ash and into the afternoon as heat and light, and a sufficiently patient accountant could in principle have followed all of it. Nothing was leaving. What was being lost was something else, something that the ledger of matter does not record: the arrangement. The particular, improbable, temporary organisation that had made those marks readable as his father's hand.

This struck him as the more honest way to state the old consolation. It is too much to say that nothing disappears, because plainly things do: his father had disappeared, and the shorthand had, and the ability to tune a piano by chest rather than by meter had gone out of the world along with the two hands that held it. What survives is the material, which does not care, and the consequence, which cannot be recalled. The arrangement is the mortal part. Every object, every man, is a knot that holds for a while in something that does not stop moving.

And yet the consequences did not go into the brazier with the paper. He could trace them without effort, sitting there in the smoke: a girl in a full hall weeping at a passage the score did not account for, and whatever she did afterward on the strength of having been moved; a sister who had built a discipline of doubt on a habit learned from a man who distrusted certainty; his

own hands, which knew wood the way the old man's had known wire, and the apprentices those hands had taught, and whatever they would go on to make after he was ash himself.

He had believed as a younger man that the indifference of the universe emptied a life of meaning, and he had come to think this exactly inverted. A thing does not require the attention of the whole to matter within the part. The galaxy does not need to notice a painting for the man standing in front of it to be altered by it. If no meaning is issued from above, then meaning is manufactured locally, at close range, between one arrangement and the next — which is not a consolation prize. It is the only kind there has ever been, including in all the centuries when men believed otherwise.

He put the last of it in near dark and swept the tray out onto the roses, where it did some good, and went inside, and the papers were gone, and nothing was gone.